

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation
(Level 1) Question Count: 5

Total Marks: 25

Code of Conduct

I MANU NEETHI S (192521063) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: Manu neethi s ____

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services.,
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

CO1 - AT1

SHORT ANSWER

Name : MANU NEETHI S

Reg. No: 192521063

Course Name: Cloud Computing and Big Data Analysis

Course code: CSA1507

Date : 28.07.2026

1. Cloud computing: It provides the service according to the pay-use demand. If you want a storage base for storing your files, you can pay according to your need and get the storage.

* It delivers computing resources (or) services such as storage, databases, networking and softwares.

* Instead of buying and maintaining physical data centres, user rent the access to these resources from providers like Amazon web, Microsoft Azure (or) Google cloud.

CHARACTERISTICS OF CLOUD COMPUTING

i) on-demand self services: you can buy the services according to your need without any direct contact.

ii) Broad network access: we can access the cloud services with our networking gadgets such as smartphones, laptops etc...

eg: If a user access services through laptop in his organisations, he can access it through mobile phones.

iii) Rapid elasticity: If a person uses the cloud platform for his business during peak festival seasons, the number

4. * It enables users to use more computing waves exponentially

ii) Networking:

- * Earlier days cards are used for network
- * As moving onto years the computers are modernised with into subject computer networks
- * Networks made communication easier between computers

iii) Distributed computing

- * Through the network the computer can share data. The task for a single user can be divided among multiple users

iv) Internet:

- * Internet is a global connectivity which connects users across the world to share files and data

v) Internet as a software.

- * The evolution of Internet makes application common among users for data sharing globally

vi) Modern computing

- * It makes virtualisation runs without pay-use demand.

Q) customers might be increased. So, in that situation he can increase the cloud service. According to the sales without crashing and after that he can reduce the cloud services.

Q) measured services:

There are four types of measured services

- * Infrastructure as a service (IaaS)

- * Platform as a service (PaaS)

- * Software as a service (SaaS)

- * anything as a service (XaaS)

2) Evolution of cloud computing

The modern computing doesn't evolved suddenly. It tooks many years to evolve and undergone various changes and improvements across the years.

Q) Hardware evolution:

In earlier days the computing hardware costs are high. The microchips are macro in size and make computing services complex.

As it evolves the hardware become small with large volume of memory and that are available in low cost.

ROLE OF DATA CENTRES IN DELIVERING CLOUD SERVICES

Data centers:

Datacentres are system that are House servers, storage system, Networking equipments and security Infrastructures That are required for Delivering cloud services

ROLES:

- * It Stores and processes Large amount of data
- * It Hosts cloud applications and services
- * It provides availability and reliability through Back^{up} system
- * It ensures Data integrity and security by encryption
- * It is helpful in providing services by adding more services when needed.

5. Differentiate Iaas, Paas, Saas, xaas

Model	Uses	Examples
Iaas (Infrastructure as a Server)	It provide storage, Networking and Data security and integrity	Amazon

Paas (Platform as a server)	It provides a platform for developing and deployment of Applications	Google search Engine
Saas (Software as a server)	It delivers software application over the Internet	Google Docs
Xaas (Anything as a server)	It provide service in cloud platform through IT Domain	Microsoft 365 (cloud service and applications)

3. Role of server virtualisation.

Virtualisation allows one physical server to run multiple virtual machines. Each with its own operating system and applications. It improve Hardware Utilization, reduce costs, and enables efficient resource management.

VIRTUALISATION	CLOUD COMPUTING
* Technology that creates virtual machines	Service model that delivers computing resources over the Internet.
* Run multiple operating system on one physical server	* Provide services like IaaS, PaaS, SaaS, XaaS
* Focuses on efficient Hardware utilization	Focuses on on-demand service delivery

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow